

SECTION A: CLOUD COMPUTING PROVIDES ON-DEMAND, ELASTIC, AND MEASURED IT RESOURCES

SCENARIO QUESTION 1: Fintech Startup in Kampala — On-Demand IT Resources (20 marks)

Background:

A fintech startup in Kampala, Uganda, wants to offer instant micro-loans to market vendors who have no formal credit history. The startup has access to mobile money transaction data, airtime purchase patterns, and social network connections. They need to make loan decisions in under 30 seconds, comply with Bank of Uganda regulations, and keep costs low. The founders want to understand how on-demand cloud resources can help them compete with established banks.

Questions:

- a) Define "on-demand self-service" in the context of cloud computing and explain how it benefits this fintech startup. (4 marks)
 - b) Explain how the startup could provision additional computing resources automatically when loan applications surge without waiting for human intervention from the cloud provider. (4 marks)
 - c) Describe TWO business scenarios where on-demand resource provisioning would give this startup a competitive advantage over traditional banks. (4 marks)
 - d) Discuss the potential risks of relying entirely on on-demand provisioning without proper cost monitoring. (4 marks)
 - e) Compare the on-demand model with the traditional procurement model that a bank would use to acquire servers, and explain why the on-demand model is better suited for a startup. (4 marks)
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SCENARIO QUESTION 2: E-Learning Platform in Kenya — Elastic IT Resources (20 marks)

Background:

An e-learning platform in Kenya serves 200,000 students during normal periods but experiences a 500% increase in traffic during national examination periods. The platform hosts video lectures, interactive quizzes, and live tutorials. The company wants to ensure smooth performance during peak periods without paying for excess capacity during off-peak periods.

Questions:

- a) Define "rapid elasticity" in cloud computing and explain how it applies to this e-learning platform. (4 marks)
- b) Explain how the platform could automatically scale out during examination periods and scale back during off-peak periods. (4 marks)

- c) Describe TWO challenges that the platform might face when scaling rapidly and propose solutions for each. (4 marks)
 - d) Discuss how elasticity differs from scalability, and explain why both concepts are important for this platform. (4 marks)
 - e) Calculate the potential cost savings if the platform pays for 200,000 users during 10 months but scales to 1.2 million users during 2 examination months, compared to maintaining capacity for 1.2 million users year-round. (4 marks)
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SCENARIO QUESTION 3: Hospital Management System in Tanzania — Measured IT Resources (20 marks)

Background:

A hospital management system in Tanzania serves 50 hospitals across the country. The system handles patient records, appointment scheduling, billing, and inventory management. Usage varies significantly: high during weekday daytime hours, low during nights and weekends. The hospital network wants to understand how measured service can help them manage costs fairly across all hospitals.

Questions:

- a) Define "measured service" in cloud computing and explain how it works like a utility meter. (4 marks)
- b) Explain how the hospital network could use measured service to allocate costs fairly among the 50 hospitals based on their actual usage. (4 marks)
- c) Describe TWO metrics that the cloud provider might measure to bill the hospital network accurately. (4 marks)
- d) Discuss how measured service could encourage hospitals to optimise their usage and reduce waste. (4 marks)
- e) Explain how the pay-per-use model of measured service differs from traditional fixed-cost IT contracts and why this matters for a network of 50 hospitals with varying needs. (4 marks)